

Class #3:

Related Terms

 **Dense Layer** - A dense layer (also called a fully connected layer) is a fundamental building block in artificial neural networks (ANNs). It is widely used in architectures such as Feedforward Neural Networks (FNNs), Convolutional Neural Networks (CNNs), and Transformers.

 **Convolutional** - In Artificial Intelligence (AI), the term "convolutional" refers to the mathematical operation of convolution, which is extensively used in Convolutional Neural Networks (CNNs) for processing images, videos, and other spatially structured data.

 **CNN** - A Convolutional Neural Network (CNN) is a type of deep learning model specifically designed for processing structured grid data, such as images and videos. CNNs are widely used in computer vision tasks like image classification, object detection, facial recognition, and medical image analysis.

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Edge Detection Matrix

In AI and image processing, edge detection is a fundamental technique used to identify boundaries and object structures within an image. Matrix dot products (convolutions) play a crucial role in detecting edges.

Vertical Edge Detection

$$\begin{array}{ccc|ccc} -1 & 0 & 1 & & 1 & 0 & -1 \\ -1 & 0 & 1 & & 1 & 0 & -1 \\ -1 & 0 & 1 & & 1 & 0 & -1 \end{array}$$

Horizontal Edge Detection

$$\begin{array}{ccc|ccc} -1 & -1 & -1 & & 1 & 1 & 1 \\ 0 & 0 & 0 & & 0 & 0 & 0 \\ 1 & 1 & 1 & & -1 & -1 & -1 \end{array}$$

 **Filter/Kernel** - A filter (kernel) in a Convolutional Neural Network (CNN) is a small matrix used to extract important features such as edges, textures, and patterns from an image. Filters slide over the input image and perform convolution operations, transforming the input data into a more useful representation for classification, detection, or segmentation.

 **Stride** - In Convolutional Neural Networks (CNNs), stride refers to the step size at which the filter (kernel) moves across the input image during the convolution operation. It determines how much the filter shifts at each step and influences the output size of the feature map.

 **Dimensionality Reduction** - In computer vision, dimensionality reduction can be understood as a process that reduces the resolution or complexity of an image while preserving important structural information. Blurring is one such technique that reduces dimensionality by smoothing high-frequency details, such as edges and textures, leading to a more simplified representation.

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Auto Encoder

 **Latent Space** - In encoder-decoder architectures, the latent space (also called bottleneck representation or embedding space) is the intermediate compressed representation of input data, capturing its most essential features before reconstruction or transformation by the decoder.

Channel in Image

In computer vision and image processing, a channel represents a separate component of an image that stores intensity values for a specific colour or feature. The number of channels in an image depends on its colour model and depth.

 **Grayscale** - Single-Channel, Pixel values range from 0 (black) to 255 (white) (for 8-bit images).

 **RGB** - 3 channels, Red (R), Green (G), and Blue (B). Each pixel is a combination of the three colour intensities.

 **RGBA** - 4 Channels, includes an Alpha (A) channel for transparency. Used in images requiring transparency effects.

 **CMYK** - 4 Channels, Cyan (C), Magenta (M), Yellow (Y), and Black (K). Used in printing applications. [Join @hmaracollege for more free courses](#) 😊

 **Multi-Channel** - Some images (e.g., hyperspectral images) have dozens to hundreds of channels. Used in scientific imaging, medical scans, and satellite images.

Formula and Notations:

 The general formula to calculate the result of a single neuron in a layer is:

$$z = (w_1 * x_1 + w_2 * x_2 + \dots + w_n * x_n) + b$$

Where:

z: The weighted sum of the inputs plus the bias. This is the value that will be passed to the activation function.

w1, w2, ..., wn: The weights connecting the neuron to the inputs from the previous layer.

x1, x2, ..., xn: The inputs from the previous layer.

b: The bias term.

In matrix form:

$$Z = W * X + B$$

Where:

Z = Weighted sum (before activation)

W = Weight matrix

X = Input vector

B = Bias vector

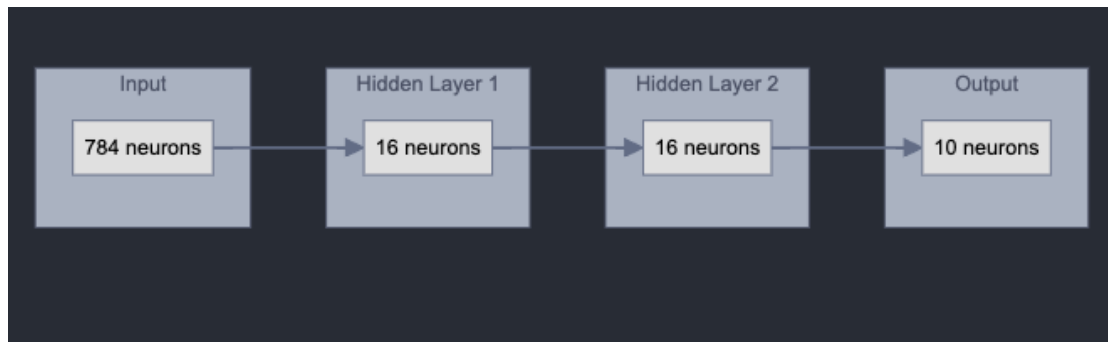
Calculation of Number of Parameters:

Consider a layer with:

1 input layer with 784 neurons (from an image of 28x28 pixels)

2 hidden layers with 16 neurons each

1 output layer with 10 neurons



Weights: (neurons of input layer x neurons of hidden layer 1 + neurons of hidden layer 1 x neurons of hidden layer 2 + neurons of hidden layer 2 x neurons of output layer)

$$= (784 \times 16 + 16 \times 16 + 16 \times 10) = 12544 + 256 + 160 = 12960$$

Bias: (16 + 16 + 10) = 42

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Total number of parameters : $12960 + 42 = 13002$

weights and biases in a convolutional neural network (CNN) :

In ANN we have **weight and biases** but in CNN we have **filters and biases**.

If you have a 3×3 filter:

– Weights: 9 parameters (3×3)

– Bias: 1 parameter

Total parameters per filter = 9 + 1 = 10 parameters

So if you have 32 filters in a convolutional layer:

- Total weights = $32 \times (3 \times 3) = 288$
- Total biases = $32 \times 1 = 32$
- Total parameters = $288 + 32 = 320$

Key point:

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Unlike dense layers where each neuron has its own bias, in convolutional layers, each filter shares a single bias term across all spatial locations.